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Research Article

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Effect of C oncentration of Mo on the Mechanical behavior of γ UMo: an Atomistic Study

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Abstract

We performed molecular dynamics simulation on nanoindentation ofγ phase Uranium Molybdenum alloys using spherical indenter. A ternary potential developed for UMoXe was utilized. We calculated the updated values for hardness and reduced elastic modulus at different concentrations of Mo. The whole process of deformation and dislocation analysis was visualized using OVITO. We found an increase in deformation withincreasein stress while dislocations are not found anyhow induced defects have been distributed throughout the simulation cell randomly. The increase in concentration affected the hardness and reduced elastic modulus significantly. This study provides insights into the structure and mechanical characteristics of γ UMo under deformation.

Keywords: Nanoindentation, MD Simulation, UMo Fuel, Spherical indenter

Introduction

The element uranium is an actinide exhibiting delocalized f electrons,exists in three solid allotropic forms: α(face-centered orthorhombic), β(body-centered tetragonal) and γ(body-centered cubic)[1]. At high temperatures, uranium transforms from αto βat approximately 935 K and β transforms to γ at approximately1045 K [2]. Several transition metals, particularly 4d and 5d elements in Group IV, through VIII, form solid solutions with γ-U, and this cubic phase can be retained in its metastable state upon cooling. It was early recognized that Mo, which has substantial solubility in U (~35%) presents a good compromise between the amount needed to stabilize γ-U and acceptable U density so achieved [3, 4]. Low Enriched Uranium (LEU) fuels play an indispensable role in peaceful purposes of nuclear energy. γUranium Molybdenum (UMo) is a LEU alloy which is considering an attractive fuel for fast and pressurized water reactors. It is ease in fabrication with inviting characteristics such as high thermal conductivity, good structural stability and reliable mechanical strength [5, 6]. Understanding the integrity during operations whether normal or transient or anticipated faults is of high importance for the better functionality of a nuclear fuel. The fission products are only released from a fuel plate during irradiation if a defect exists in the cladding that provides a path for fission gas to escape [7]. These defects under stress and temperature fields eventually lead to swelling or pores which ultimately either limit the performance or permanent failure of the fuel [8]. A group of researchers on experimental basis showed that the brittleness and fracture aredependent on Mo concentration [9]. In this work we performed a number of MD simulationsusing LAMMPS code to investigate the impact on the nanomechanical properties of variable Mo concentration in γUMo.A spherical indenter of defined radius was used and its depth with varying load was analyzed.Dislocations creation, grain formation and segregation were tracked using OVITO.

Method

MD simulations were performedusing Large scale Atomic/ Molecular Massively Parallel Simulator (LAMMPS) [10]. The EAMpotential developed by Smirnova [11] is used in this work. Periodic boundary conditions were applied in the x and y while shrink style was used along the z axis. A simulation cell of 129600 atoms was created. The relaxation was conducted using the NVT ensemble at 300 K using the velocity Verlet algorithm at 300 K with a time step of 1 fs for 100,000 steps. The system was relaxed again at 100 ps for the same number of steps using NVE ensemble [12].Using Langevin's thermostat [13] we kept the temperature constant at 300 K. A rigid spherical indenter was used with radius of 35 A0, and the impact of force on each atom was tracked. A non atomic repulsive sphere exerts a force on the atomic layer which